

Transformer u(t) Experiment Report

Manuscript time-dependent control trained by PMP/KKT optimality gaps

This report reproduces the manuscript's time-dependent Transformer control strategy $u_{\text{theta}}(t)$: $[0, T] \rightarrow [0, u_{\text{max}}]$.

1. Model and Training Objective

We use the population dynamics and cost from the manuscript:

$$\dot{N}_i(t) = (r_i - \phi_i u(t) - M_i G(N(t))) N_i(t) \quad (1)$$

$$G(N) = \log \left(1 + \frac{1}{m} \sum_{k=1}^m N_k \right) \quad (2)$$

$$J(u) = \alpha^\top N(T) + \int_0^T (\beta^\top N(t) + \gamma u(t)) dt \quad (3)$$

Reported parameters: $T=10$, $m=21$, $u_{\text{max}}=3$, $\alpha=1$, $\beta=0.1$, $\gamma=20$, and $N_i(0)=10$. The control is represented by a small Transformer encoder over normalized time.

Given $u_{\text{theta}}(t)$, we roll out $N_{\text{theta}}(t)$, solve the costate equation backward, and compute the switching function

$$\psi(t) = H_u(N, \lambda, u) = \gamma - \sum_i \phi_i \lambda_i(t) N_i(t) \quad (4)$$

component	role
non-singular KKT loss	enforces $u=0$ when $\psi>0$ and $u=u_{\text{max}}$ when $\psi<0$
singular loss	near $\psi=0$, matches $u_{\text{theta}}(t)$ to the singular candidate $u_{\text{sing}}(N(t))$

Here $q(t)$ is a smooth weight that switches between the singular condition near $\psi(t)=0$ and the boundary KKT condition away from $\psi(t)=0$.

2. Learned $u(t)$, $N(t)$, and $N(u)$

The learned control starts high, transitions to a lower interior/singular-like region, and increases again near the terminal portion.

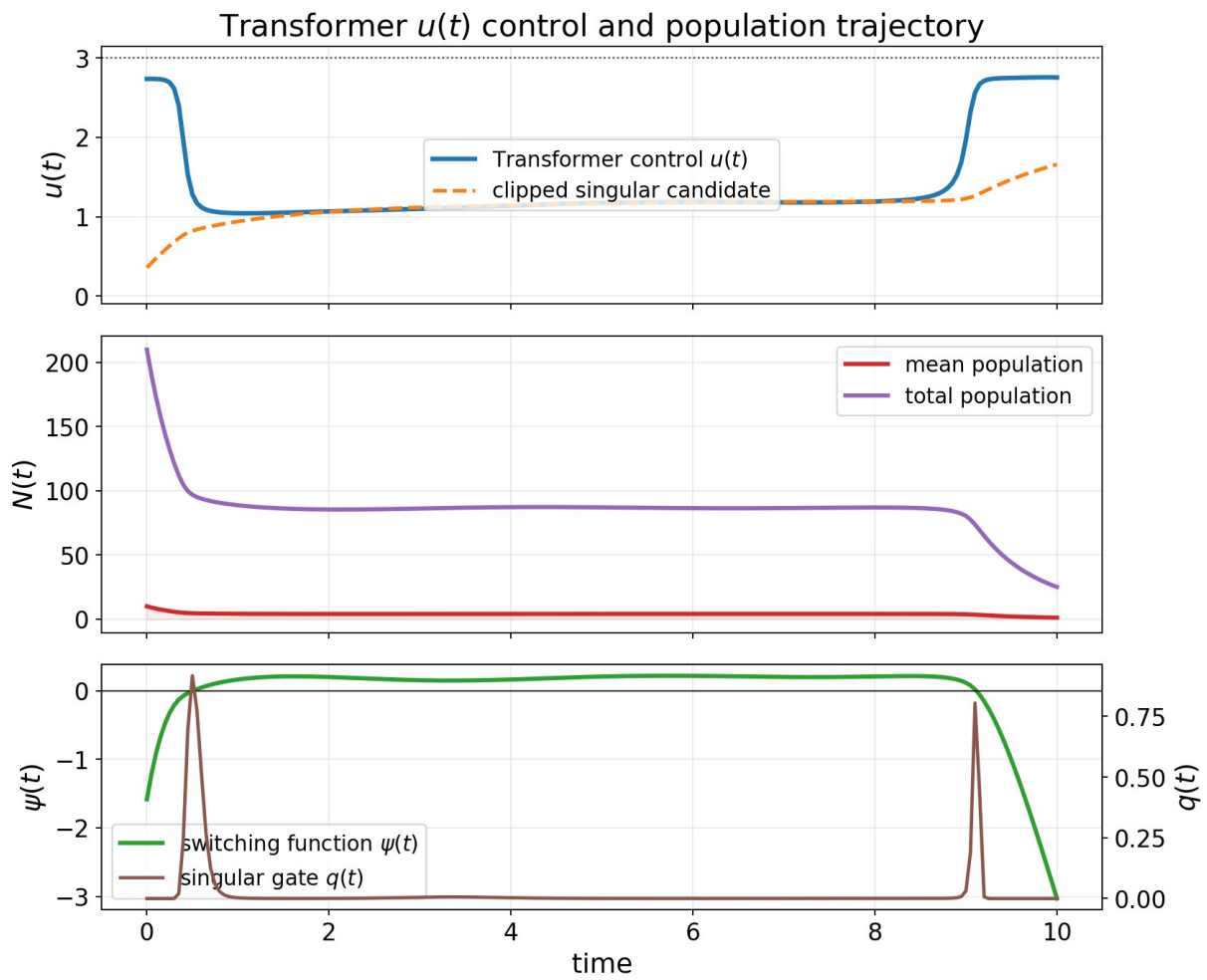


Figure 1. Learned Transformer control $u(t)$, population trajectory $N(t)$, switching function $\psi(t)$, and singular weight $q(t)$.

The $N(u)$ phase plot below uses the total population across all 21 subpopulations.

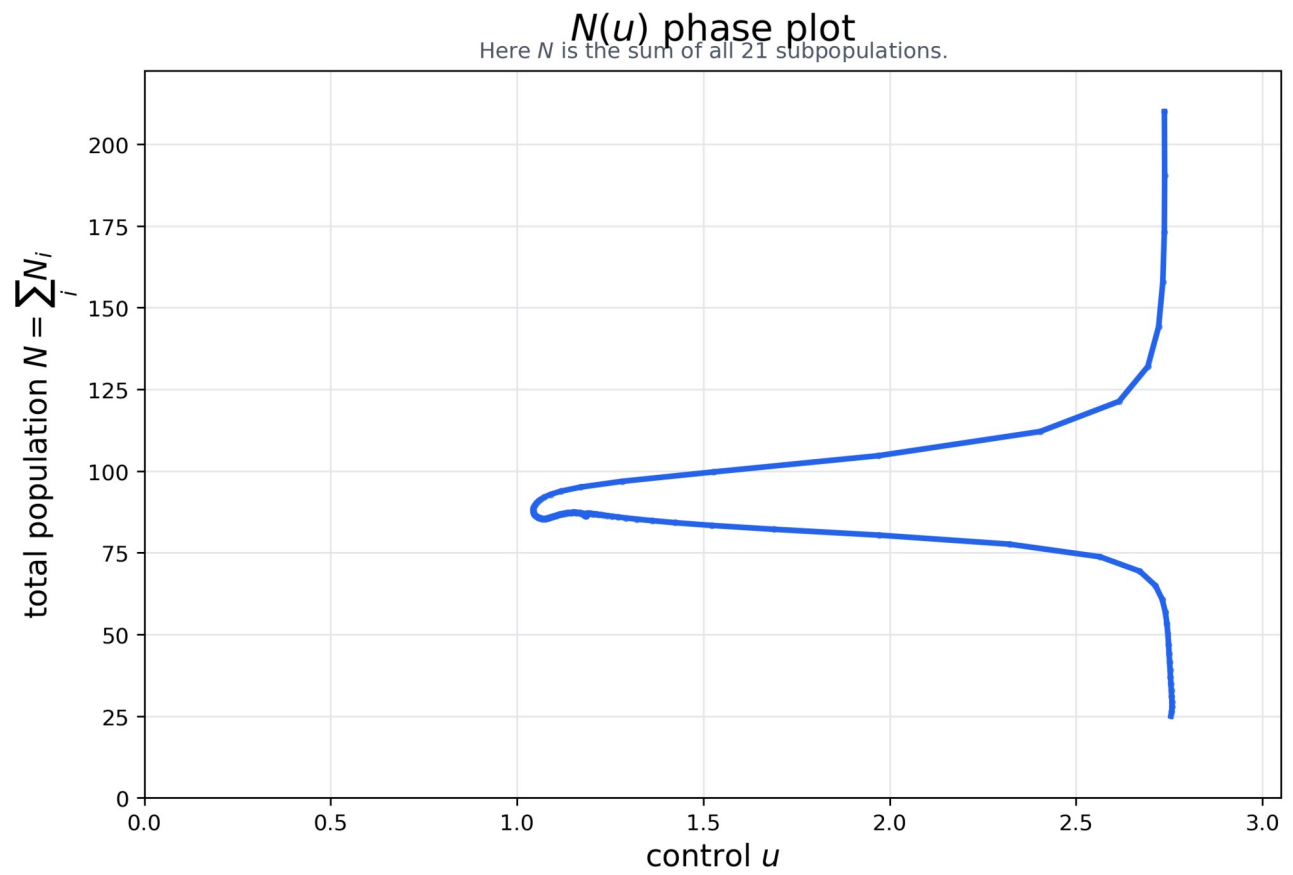


Figure 2. $N(u)$ phase plot using the total population across the 21 subpopulations.

3. Training Loss Trajectories for PMP/KKT Conditions

The table reports the smallest recorded training loss. In this experiment, the training loss is the manuscript's PMP/KKT optimality gap, composed of the singular condition and the non-singular Hamiltonian minimization condition.

metric	value
total training loss / PMP-KKT optimality gap	0.02637
singular-condition training loss	0.00167
non-singular Hamiltonian-minimization training loss	0.02471
objective J on the training grid	384.76
range and mean of $u_{\text{theta}}(t)$	min 1.038, max 2.758, mean 1.372
terminal mean state	1.207

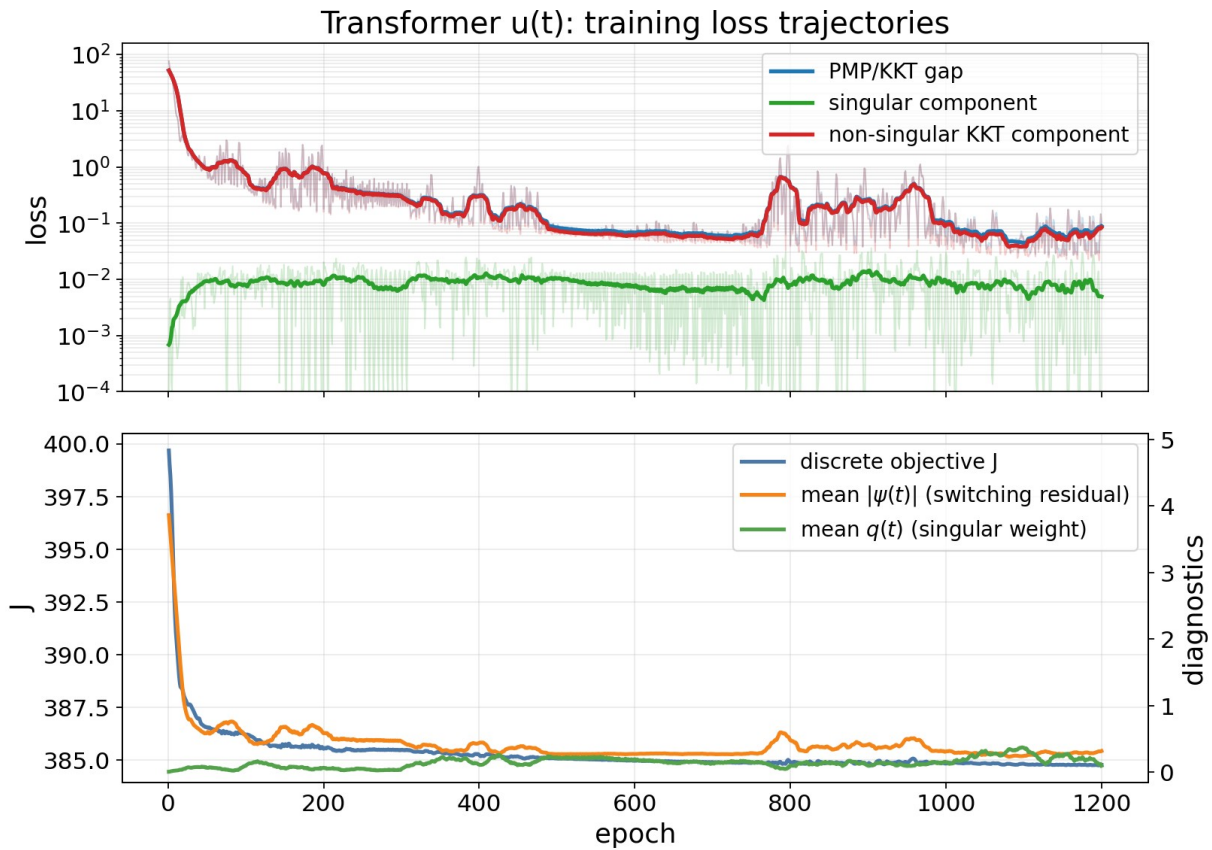


Figure 3. Training trajectories of the total PMP/KKT optimality gap and its singular and non-singular components.

The final pointwise diagnostic below uses a smooth weight $q(t)$ to separate the two regimes: near $\psi(t)=0$ it emphasizes the singular-condition error; away from $\psi(t)=0$ it emphasizes the boundary KKT error.

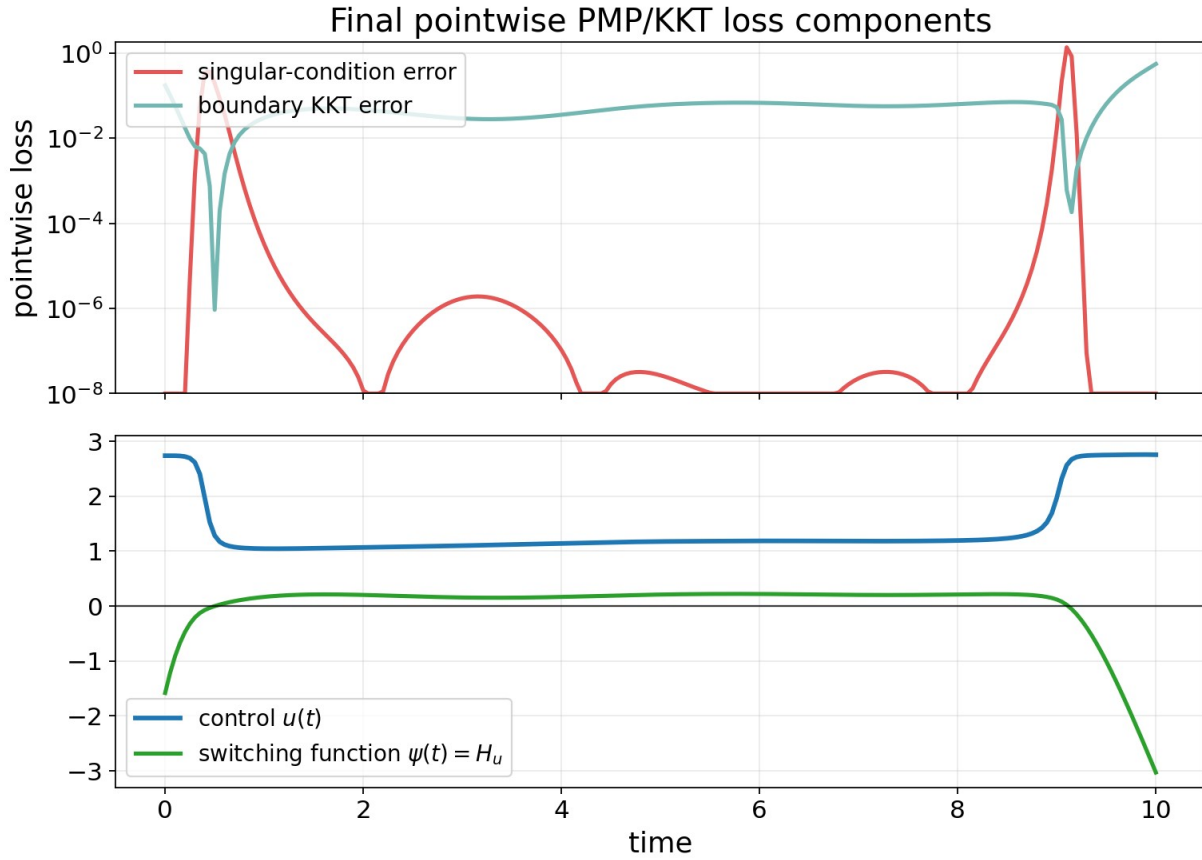


Figure 4. Pointwise singular-condition error and boundary KKT error along the final learned trajectory.

4. Comparison With Related Work

The related work in the manuscript uses several different optimal-control formulations, so we first compare them at the level of the learned object and optimality condition before giving the numerical baseline.

reference	learned object / formulation	relation to this report
HJB / BSDE methods [1,2,7]	value function $V(t,N)$ or HJB PDE solution	full state-domain feedback problem; not directly comparable to the requested time-dependent $u(t)$ reproduction
Neural-PMP [3]	control sequence via forward rollout, backward costate recursion, and Hamiltonian-gradient updates	closest related-work baseline for the present $u(t)$ experiment; implemented numerically below
DeepONet HJB policy iteration [5]	operator for value-function / HJB policy-iteration solves	methodological reference for feedback/value-function learning; requires a different training problem
PINN policy iteration [6]	policy evaluation / improvement PDEs with neural networks	feedback/value-function route, not the same objective as trajectory-level $u(t)$ PMP/KKT loss
classical chemotherapy OC [4]	PMP and singular-control structure for heterogeneous population model	source of the singular-control formula used in the manuscript, not a separate learning baseline

Among these, [3] is the closest apples-to-apples numerical comparison. Here [3] refers to Gu, Xiong, and Chen, Pontryagin Optimal Control via Neural Networks (arXiv:2212.14566). Their Neural-PMP method contains two parts: learning a differentiable dynamics model from data, and then using a PMP-gradient update to optimize the control sequence. Since the dynamics are known in our manuscript setting, we compare against an oracle-dynamics version of the second part: forward state integration, backward costate recursion, and Hamiltonian-gradient updates of a discrete control sequence. This row should therefore be read as a controller-stage baseline following [3].

method	control update / training criterion	PMP/KKT gap	objective J^*	note
Transformer $u(t)$	paper PMP/KKT optimality-gap loss	0.0523	386.70	main $u(t)$ reproduction
Neural-PMP controller-stage [3]	Hamiltonian-gradient update with known dynamics	7.47	387.02	related-work comparison
direct grid cost	minimize discretized J	0.805	386.47	reference only
constant $u=1.5$	fixed control	76.89	400.40	scale check

* The J values in this table are computed after fixing $u(t)$, reintegrating $N(t)$ with a finer-step fourth-order Runge-Kutta method, and then applying the manuscript objective definition. This is only to use the same numerical integration accuracy across methods.

Under the same PMP/KKT gap calculation, the Transformer $u_{\text{theta}}(t)$ has a much smaller gap than this Neural-PMP controller-stage baseline. When J is recomputed with the same numerical evaluator, the Transformer also has a slightly lower objective value in this run.

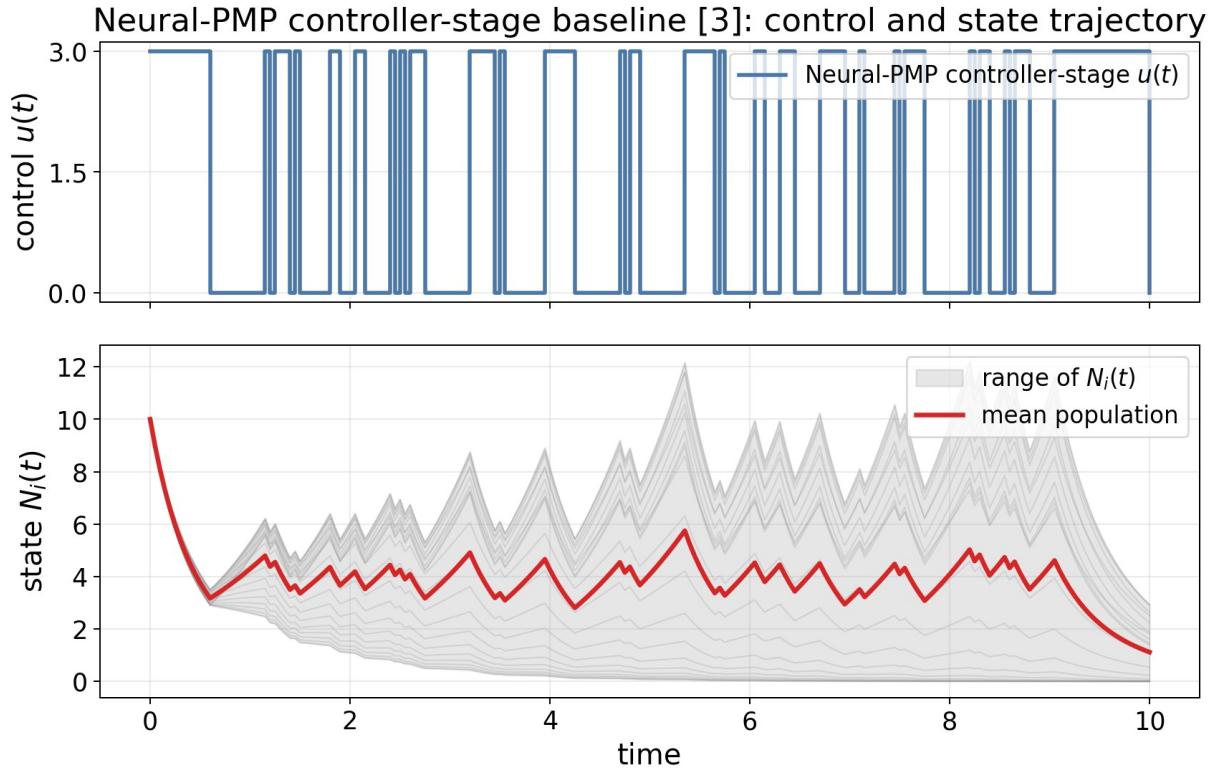


Figure 5. Neural-PMP controller-stage baseline [3]: control sequence and resulting state trajectory.

Neural-PMP controller-stage baseline [3]: selected-run training trajectories

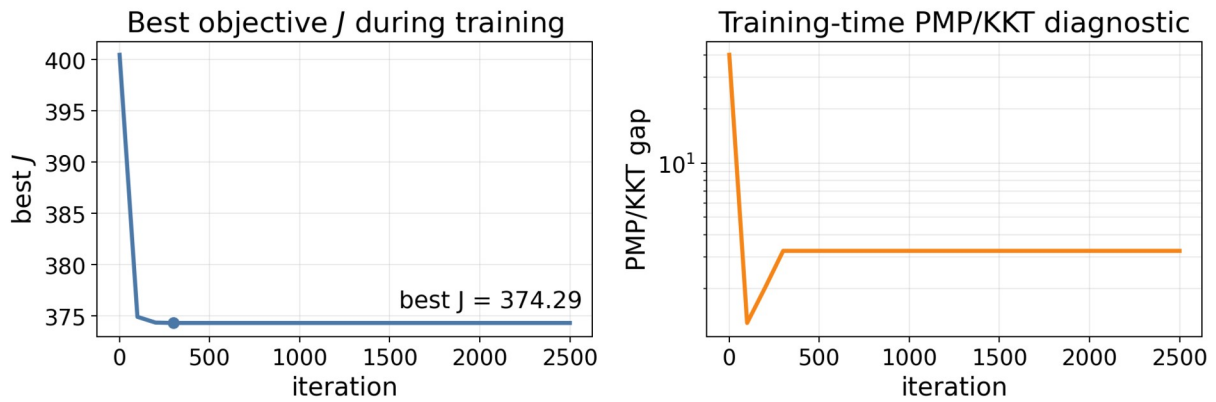


Figure 6. Neural-PMP controller-stage baseline [3]: selected-run training trajectories.

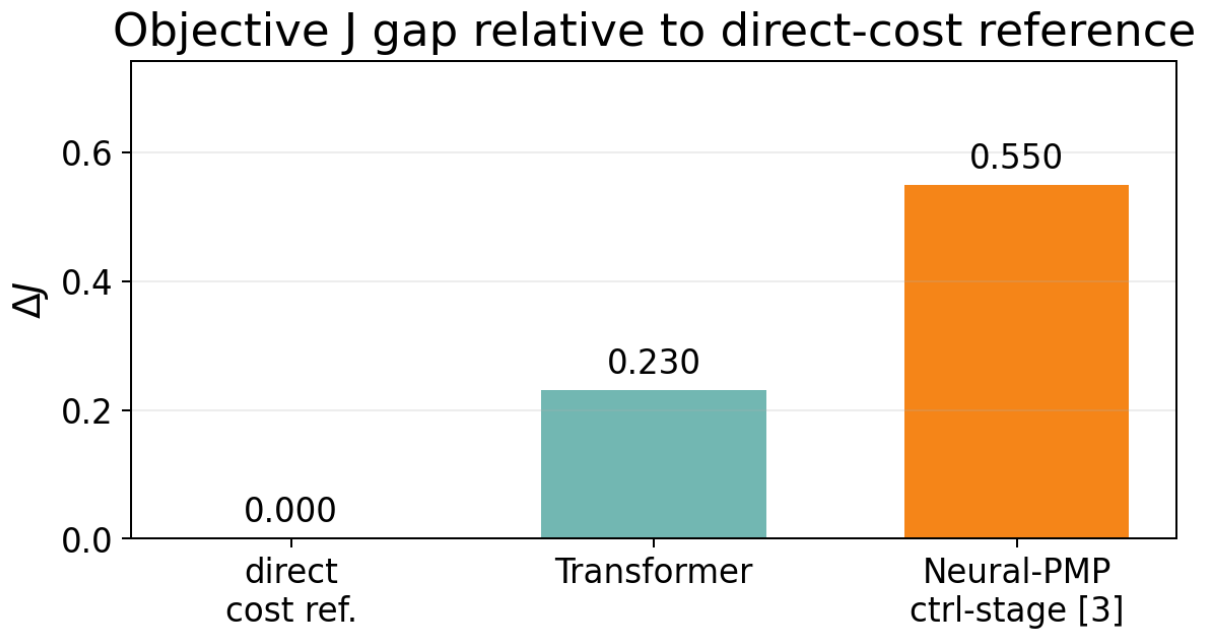


Figure 7. Objective J gap relative to the direct-cost reference under the common numerical evaluation.

5. Conclusion

The requested $u(t)$ reproduction is complete. The Transformer control trajectory trained with the manuscript's PMP/KKT optimality-gap loss is smooth and satisfies the control bounds, reduces the training optimality gap from 76.26 to 0.02637, and gives J approximately 386.70 under the common numerical evaluation. The state-dependent extension $u(t, N)$ is left for separate work because it requires different optimality conditions.